Multi-Branch Kinematic Analysis Report (Separate Branch Visualization)

Automated Generation System

March 13, 2026

1 Assembly Modes (Branches) Overview

Due to the highly nonlinear nature of the closed-loop constraints, spatial 7R mechanisms typically exhibit multiple assembly modes (kinematic branches) for the same driving input. Our algorithm successfully identified **4** distinct branches by applying diverse initial perturbations at $q_1 = 0°$.

2 Matrix Calculation (Branch A Example at $q_1 = 0°$)

We extract the matrices for the first identified branch (Branch A). The full joint configuration is:

$$\mathbf{q}_A = \begin{bmatrix} 0.0000 \\ -0.0231 \\ -0.0002 \\ 0.0462 \\ 0.0003 \\ 0.0231 \\ -0.0005 \end{bmatrix} \tag{1}$$

The passive Jacobian $J_p \in R^{6\times 6}$ (removing column 1):

$$J_p = \begin{bmatrix} 0.000 & 0.000 & 2.499 & 0.058 & -2.500 & -0.000 \\ -1.000 & -0.000 & 2.999 & 0.000 & 1.998 & -0.000 \\ 0.000 & -3.000 & -0.058 & -0.500 & -0.058 & -2.000 \\ 0.000 & 0.000 & -0.023 & 0.000 & -0.023 & -0.001 \\ 0.000 & -1.000 & -0.000 & -1.000 & -0.000 & 1.000 \\ 1.000 & 0.000 & -1.000 & -0.000 & 1.000 & -0.000 \end{bmatrix} \tag{2}$$

Singular Values matrix Σ:

$$\Sigma = \begin{bmatrix} 4.178 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 3.671 & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & 3.352 & 0.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & 1.669 & 0.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & 0.892 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 \end{bmatrix} \tag{3}$$

3 Kinematic & Singularity Plots for Each Branch

The continuous kinematic mappings and corresponding Singular Value Decomposition (SVD) evolutions for each identified branch are plotted below.

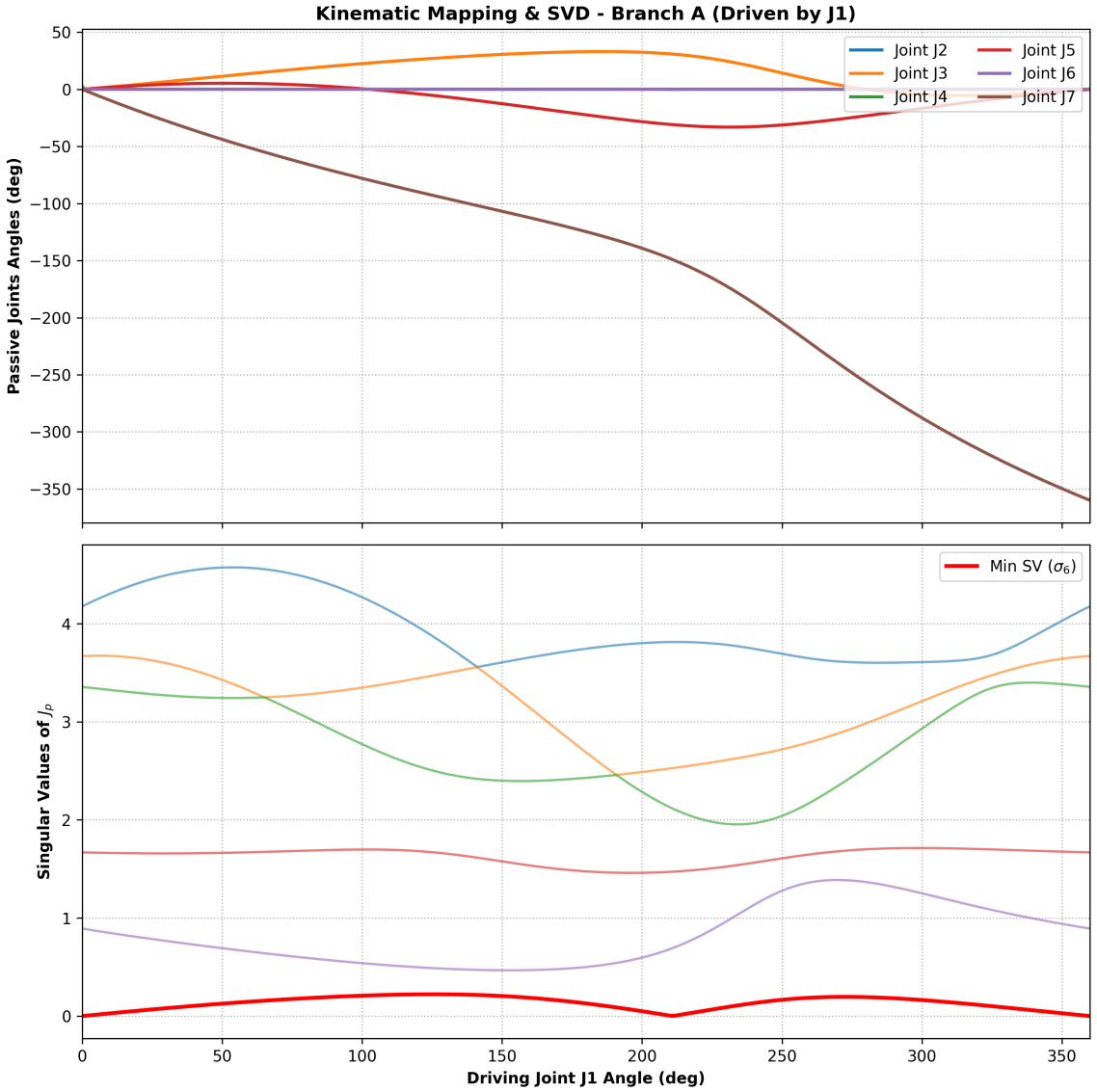


Figure 1: Kinematic mapping and singular value evolution for **Branch A** (Driven by J_1).

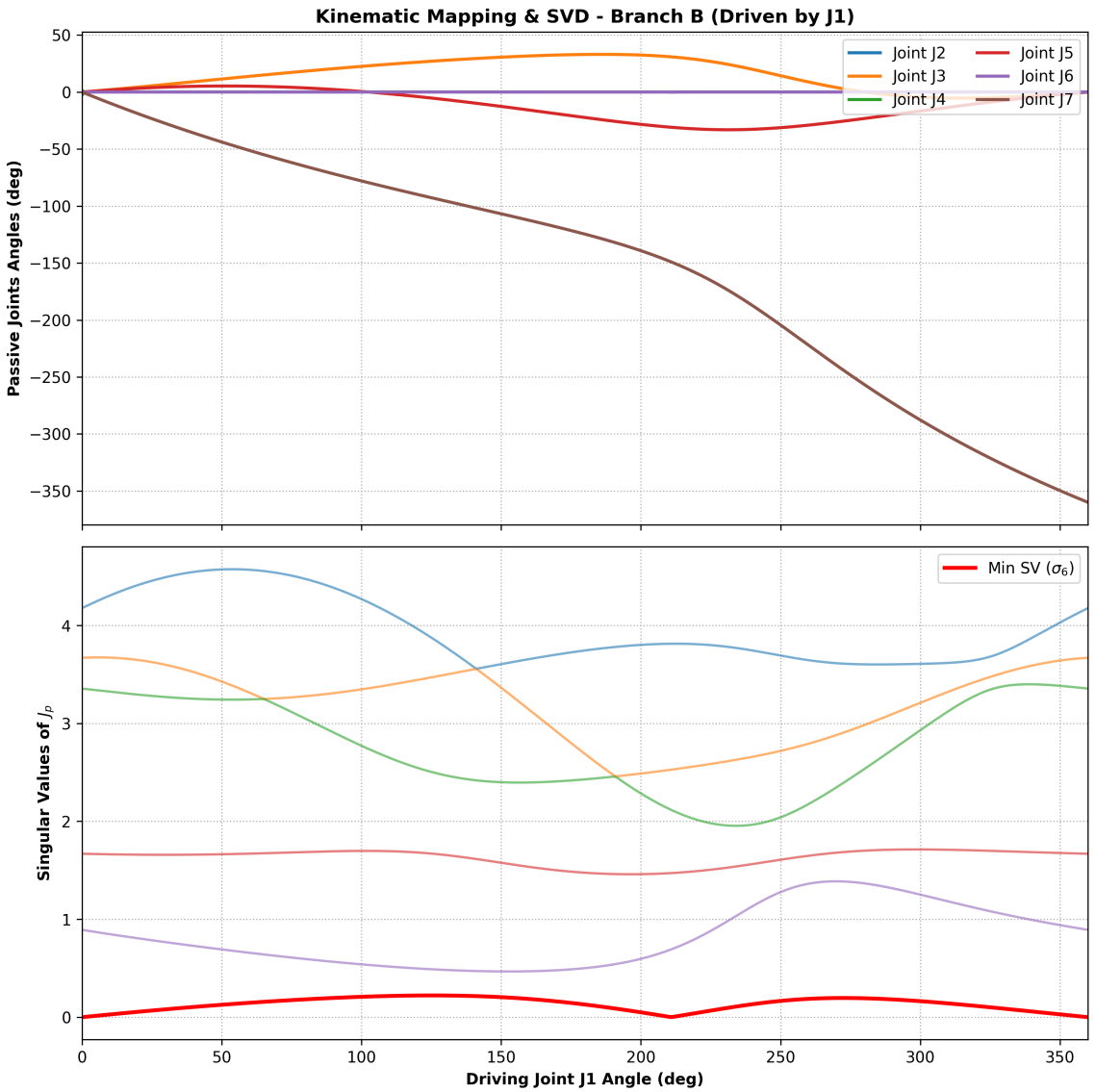


Figure 2: Kinematic mapping and singular value evolution for **Branch B** (Driven by J_1).

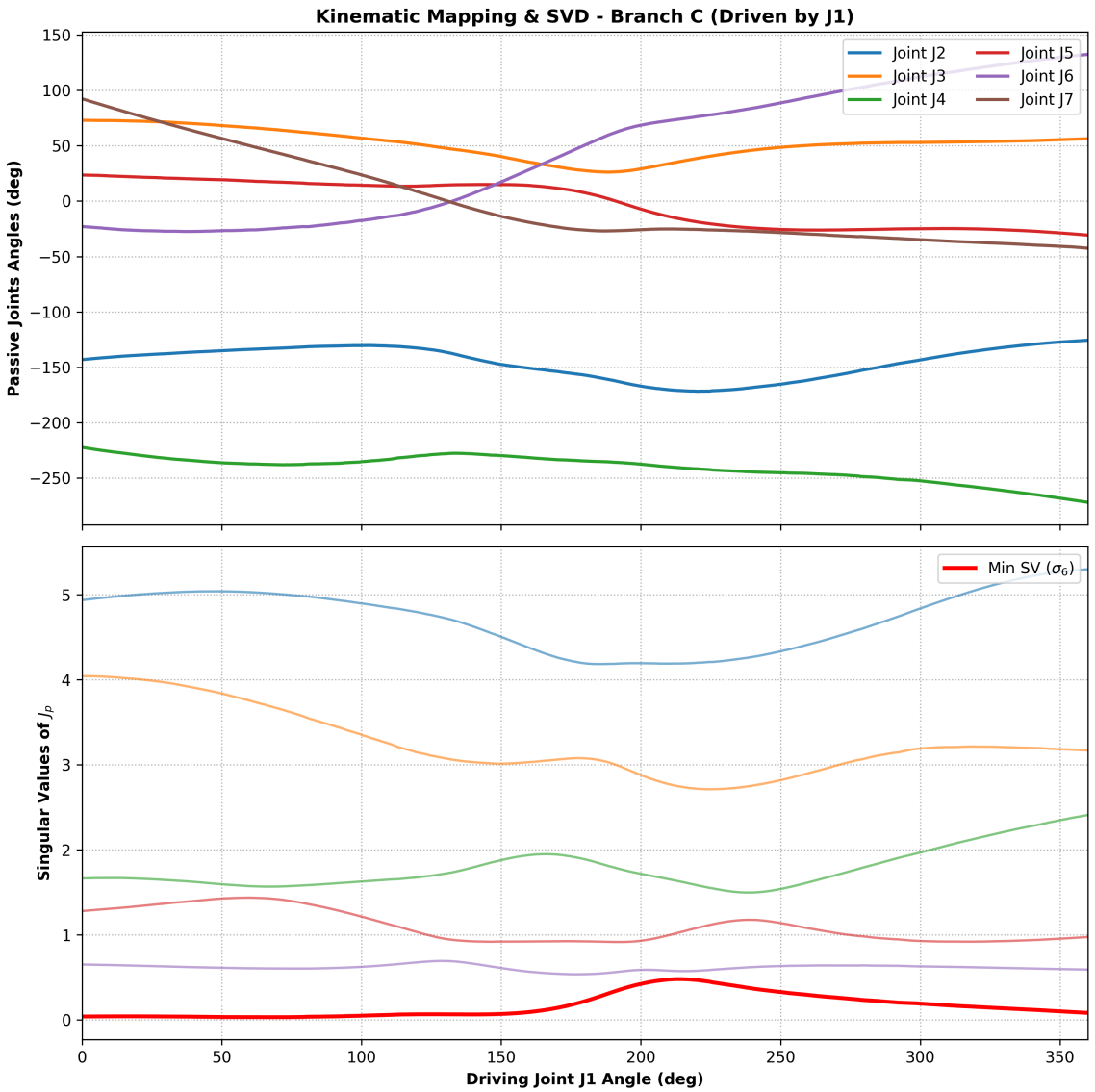


Figure 3: Kinematic mapping and singular value evolution for **Branch C** (Driven by J_1).

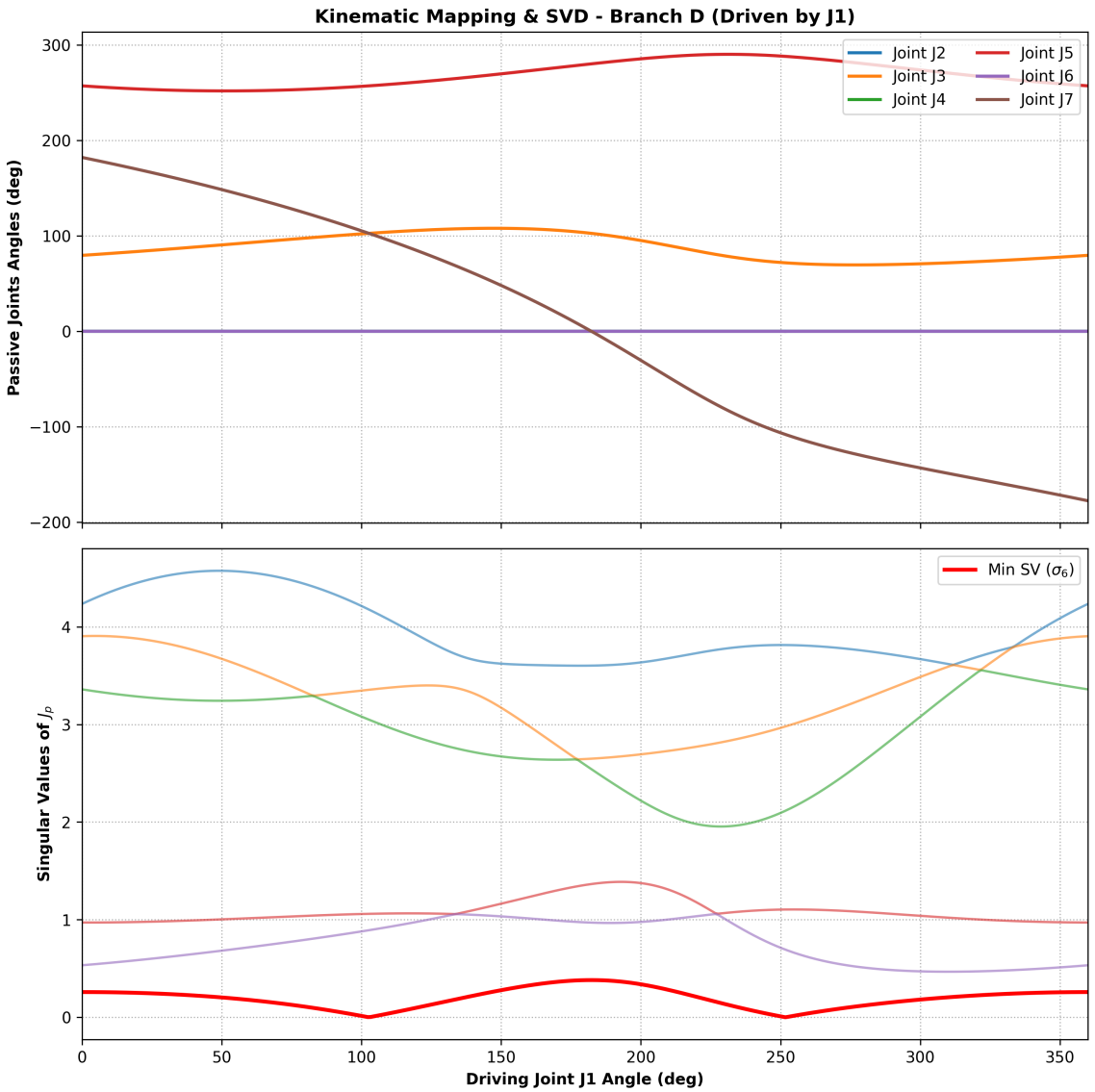


Figure 4: Kinematic mapping and singular value evolution for **Branch D** (Driven by J_1).